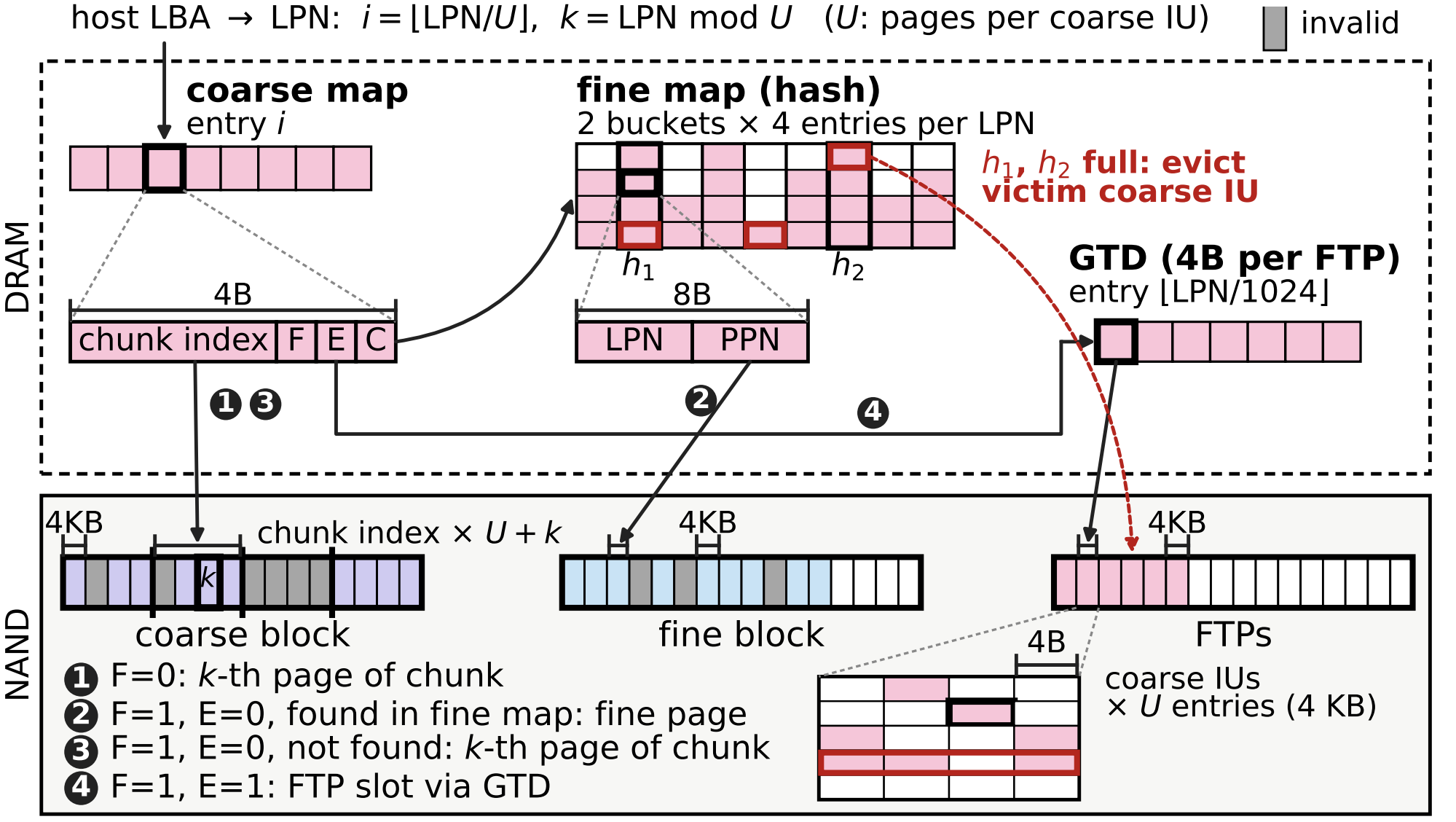
host LBA → LPN: $i = \lfloor \mathrm{LPN}/U \rfloor$, $k = \mathrm{LPN} \bmod U$ (U: pages per coarse IU)
invalid
DRAM
coarse map
entry i
4B
chunk index
F
E
C
fine map (hash)
2 buckets × 4 entries per LPN
h_1
h_2
8B
LPN
PPN
h_1, h_2 full: evict
victim coarse IU
GTD (4B per FTP)
entry ⌊LPN/1024⌋
❶ ❸
❷
❹
NAND
4KB
chunk index × $U + k$
k
coarse block
4KB
fine block
4KB
FTPs
4B
coarse IUs
× U entries (4 KB)
❶ F=0: k-th page of chunk
❷ F=1, E=0, found in fine map: fine page
❸ F=1, E=0, not found: k-th page of chunk
❹ F=1, E=1: FTP slot via GTD